Project Proposal: Smart Expense Tracker

1. Introduction

Smart Expense Tracker is a web application designed to help users manage their daily income and expenses. Users can record transactions, monitor their spending, set monthly budgets, and view their financial summaries through a simple dashboard.

2. Problem Statement

Managing daily expenses manually can be difficult and time-consuming. People may forget to record transactions or exceed their budgets. Therefore, a simple digital system is needed to manage expenses efficiently.

3. Aim

To develop a user-friendly web application that helps users manage income and expenses using CRUD operations.

4. Objectives

- To add and manage income and expense records.
- To view, update, and delete transaction details.
- To store transaction data in a MySQL database.
- To calculate total income, expenses, and remaining balance.
- To display expense summaries and charts.
- To help users monitor monthly budgets.

5. Main Features

- Add new income and expense transactions.
- View all transaction records.
- Update existing transaction details.
- Delete unwanted transactions.
- Categorize transactions, such as food, transport, education, and bills.
- Display total income, total expenses, and current balance.
- Set monthly budgets and view spending charts.

6. Technologies Used

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python FastAPI
- **Database:** MySQL
- **Database integration:** SQLAlchemy
- **Charts:** Plotly
- **Development tool:** Google Antigravity

7. CRUD Operations

- **Create:** Add new income or expense records.
- **Read:** View saved transaction records.
- **Update:** Edit existing transaction details.
- **Delete:** Remove unwanted transactions.

8. Expected Outcome

The expected outcome is a functional web application that allows users to manage their personal finances easily. The system will provide accurate transaction records, financial summaries, and budget tracking to help users understand their spending habits.

9. Conclusion

Smart Expense Tracker will provide a simple solution for personal expense management. This project will also provide practical experience in Python development, REST API creation, MySQL database management, and CRUD operations.